Multi-molecule RPSD estimation and 3D KLT picking

Findings from the TomoTwin generalization tomogram

Research summary

Experiments performed 17–18 July 2026

Abstract

This note consolidates the recent experiments on radial power spectral density (RPSD) estimation and 3D Karhunen–Loève transform (KLT) particle picking in the TomoTwin generalization tomogram. The detailed 2DF7 experiment achieved 11/12 localization at the known particle count with the finalized package pipeline. Fresh end-to-end fits at each molecule's TomoTwin evaluation box remained strongly biased toward 2DF7 when only the requested class count was retained. However, a one-pass detector supplied with the aggregated count of 51 molecules recovered 36/51 centers (70.6%) and represented all seven classes. Thus rank-one ALS produces a broadly particle-sensitive score with a strong 2DF7 bias at its highest ranks, rather than a strictly 2DF7-specific score. A constrained $K = 7$ ALS extension failed to separate molecule-specific components: oracle and shared-diameter evaluations both localized only 2DF7. Clustering higher-support RPSDs extracted at detected centers retained weak-to-moderate class information but reached only 44.4% mapped accuracy. PCA improved clustering stability, not molecular separation. Together, the experiments separate useful broad particle detection from the substantially harder task of molecular identification.

1 Data, ROI, and experimental conventions

The reconstruction has shape $200 \times 512 \times 512$ in (z, y, x) and a voxel spacing of approximately 10.204 Å. The detailed localization experiments used the 128^3 ROI beginning at $(z, y, x) = (24, 16, 304)$. Ground-truth coordinates were converted once from the stored (x, y, z) convention to array-order (z, y, x).

Table 1 combines the TomoTwin evaluation boxes, molecular masses reported for the generalization tomogram, and the number of GT centers in this ROI. "Fully supported" means that the class-specific box fits inside the ROI using a conservative $D_k/2$ boundary margin. The known-count script used every center in the ROI as its requested count; the fully-supported count is reported separately as a boundary diagnostic.

Table 1: Molecular classes in the localization ROI.

Class	Mass (kDa)	Evaluation box D_k	Centers in ROI	Fully supported
1AVO	149	18	8	7
1E9R	276	21	5	2
1FPY	632	18	4	2
1FZG	142	28	9	5
1JZ8	512	20	4	2
1OAO	315	20	9	3
2DF7	896	33	12	7
Total			51	28

The box size is a spatial-extent parameter, not a molecular-mass measurement. For example, 1FPY is 632 kDa but has an 18-voxel box, whereas the elongated 1FZG structure is only 142 kDa but has a 28-voxel box. Nevertheless, 2DF7 is the maximum under both measures and is also the most frequent class in the selected ROI.

2 Detailed 2DF7 experiment

2.1 Finalized geometry and spectral settings

The 2DF7 experiment began with the 33-voxel TomoTwin evaluation box. The PSD patch was derived from the package heuristic

$$p = \text{odd\_floor}(0.8D) = \text{odd\_floor}(26.4) = 25.$$

The final package test used the parameters in Table 2.

Table 2: Final 2DF7 package configuration.

Parameter	Value
Particle diameter setting	$D = 33$ voxels
PSD patch size	25^3 voxels
Bandpass removed at low/high ends	5%/5%
ALS maximum iterations	500
Legendre quadrature order	40
Maximum angular order	4
Fredholm radius	$0.4D = 13.2$ voxels
Template side	29 voxels
NMS radius	$0.5D = 16.5$ voxels
Matching tolerance	$0.5D = 16.5$ voxels

The first ALS pass separated a common noise spectrum from a single particle spectrum and calibrated both to the patch variances (Fig. 1). A noise-based prewhitening step was then applied to the full ROI, followed by a second ALS fit.

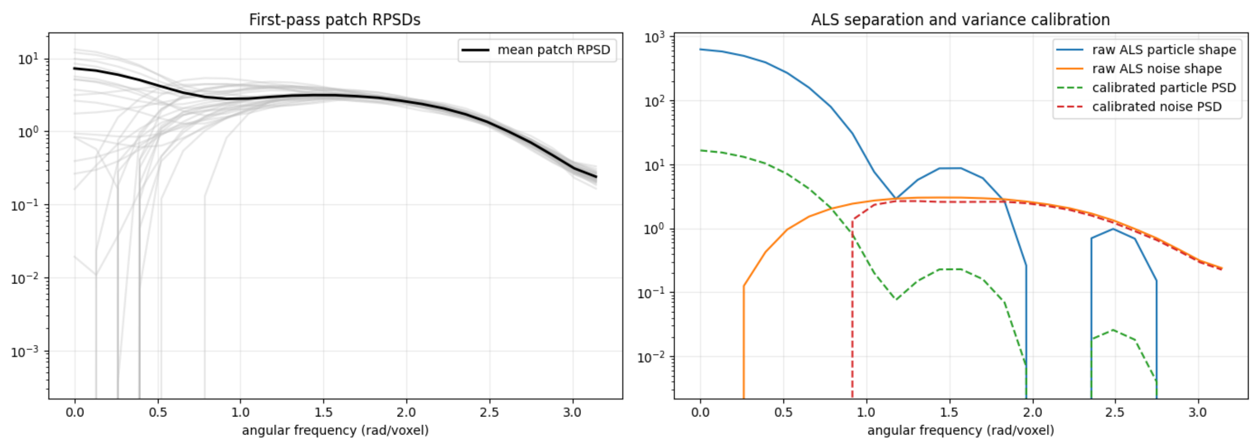


Figure 1: First-pass patch RPSDs and the calibrated rank-one ALS separation. Each gray curve is a patch RPSD; the right panel shows the fitted particle and noise spectra after variance calibration.

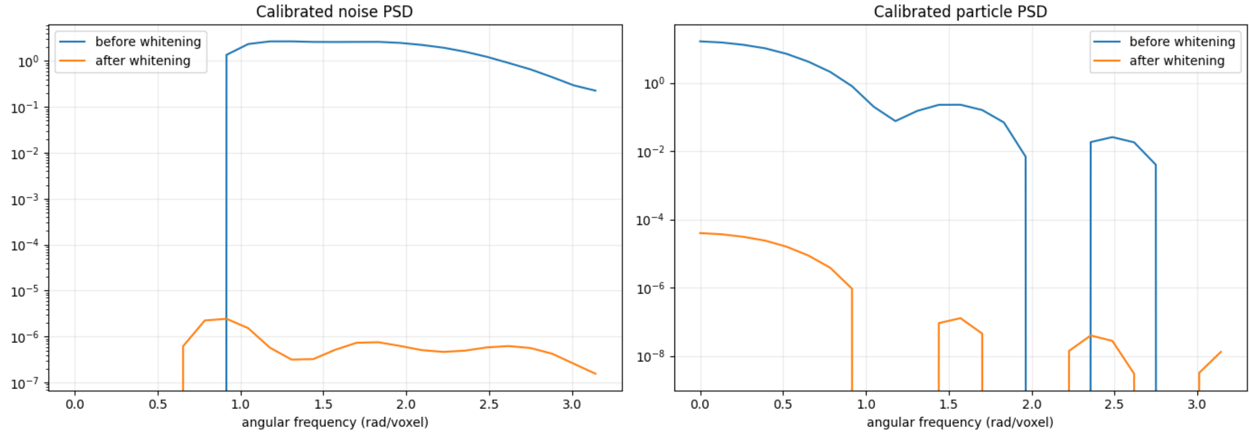


Figure 2: Calibrated noise and particle spectra before and after whitening. Whitening executed numerically, but the second-pass noise-spectrum coefficient of variation increased from 0.865 to 1.036 rather than becoming flatter. This is a diagnostic warning that the isotropic noise model and rank-one separation are imperfect on the heterogeneous tomogram.

2.2 Likelihood construction and parameter sweeps

The post-whitening particle RPSD was used in the radial Fredholm equations. The retained eigenfunctions were expanded into 3D spherical-harmonic templates, orthogonalized, and used to evaluate the Gaussian log-likelihood score throughout the valid-convolution volume. Figure 3 shows that the score is spatially concentrated but strongly heavy-tailed: the highest-scoring location is much larger than the remainder.

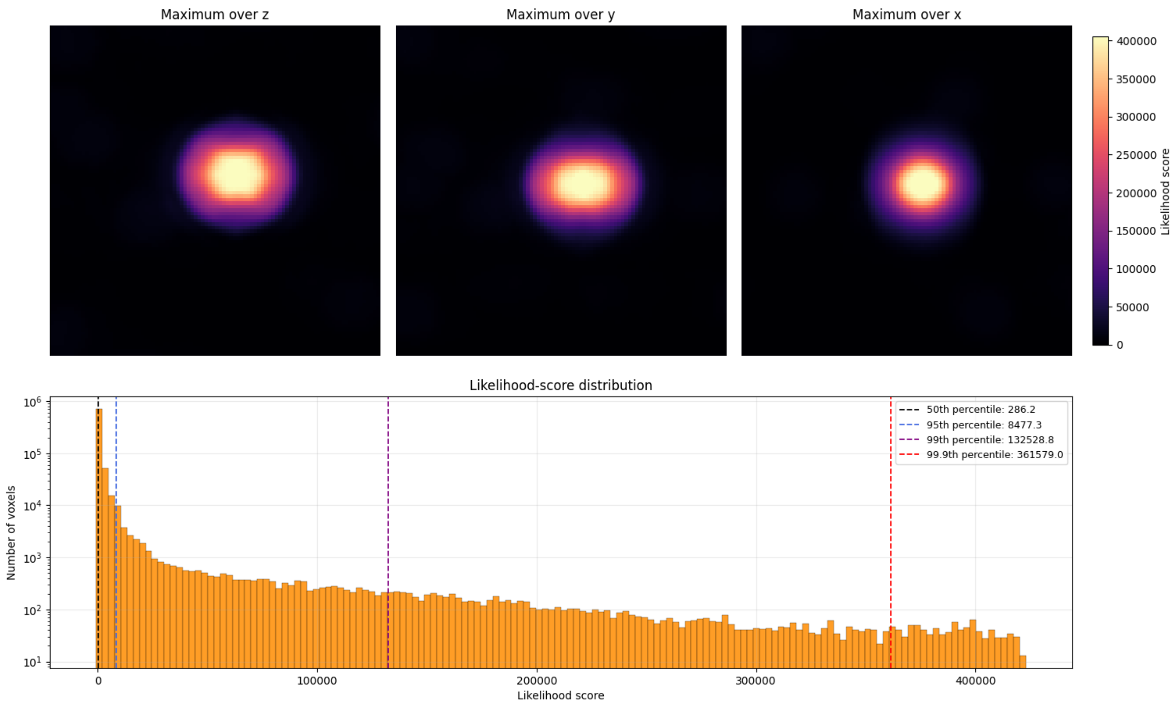
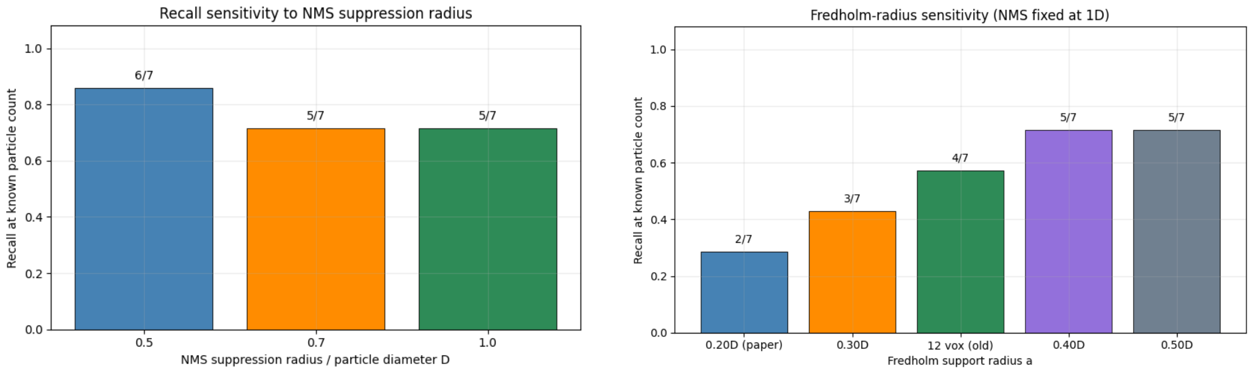


Figure 3: Maximum projections of the 3D likelihood tensor and its score distribution for the detailed 2DF7 model.

Two controlled sweeps established the finalized geometric settings. Reducing the NMS radius from $1.0D$ to $0.5D$ increased fixed-score known-count recovery from 5/7 to 6/7. Increasing the Fredholm support from the paper-scale $0.2D$ to $0.4D$ increased recovery from 2/7 to 5/7 in the manually reconstructed pipeline; $0.5D$ did not improve it further.



(a) NMS-radius sweep on a fixed score tensor.

(b) Fredholm-support sweep with fixed NMS.

Figure 4: Geometry sweeps used to select the final package settings. These manual-pipeline sweeps are diagnostic and should not be conflated with the final end-to-end package result.

2.3 Final package localization

The exact package method was finally run with the optimized support and NMS settings and with the total count of 12 supplied. It recovered 11/12 GT centers:

$$\text{recall@12} = \frac{11}{12} = 0.917.$$

All seven fully-supported centers were recovered, with a mean matched distance of 3.09 voxels and a maximum of 4.19 voxels. Four of the five ROI-boundary centers were recovered, with mean and maximum distances of 5.46 and 6.37 voxels, respectively.

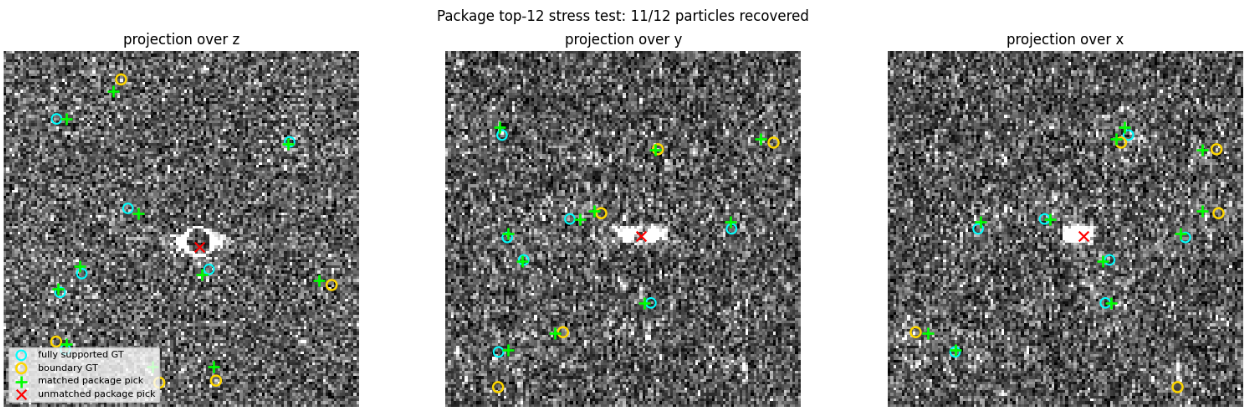


Figure 5: Final package top-12 stress test. Green symbols denote matched picks; red denotes the unmatched pick; cyan and yellow circles mark fully-supported and boundary GT centers.

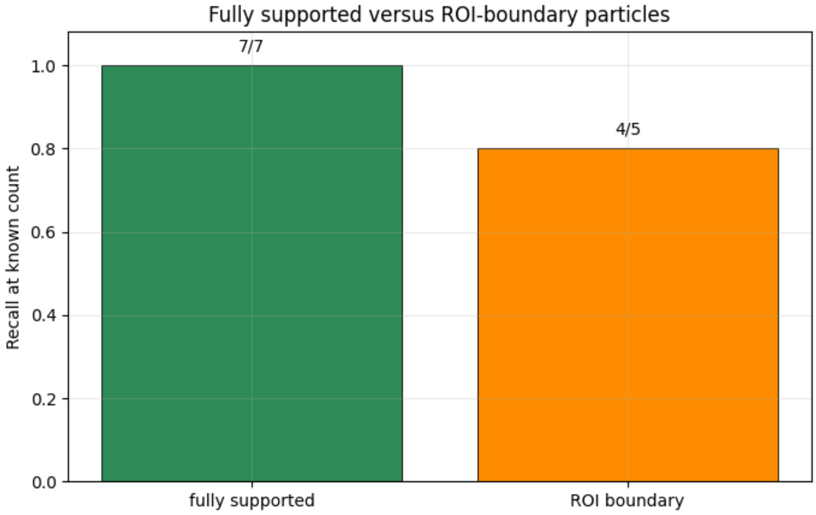


Figure 6: Final package recall separated by geometric support: 7/7 fully-supported and 4/5 ROI-boundary particles.

The successful 2DF7 result initially suggested that the estimated particle RPSD might be 2DF7-specific. The subsequent multi-molecule experiments show that this interpretation is too strong.

3 Per-molecule RPSD measurements

3.1 Common-grid comparison

To compare spectral content directly, GT-centered cubes were extracted for all seven classes using the same 33^3 crop and the same voxel grid. This common crop was deliberate: changing array length does not change the physical Nyquist frequency at a fixed voxel spacing, but it changes the discrete frequency grid and windowing. The common crop therefore isolates differences in molecular content from differences in sampling grid.

For each class, the observed periodograms were averaged over all instances. An empirical background RPSD was estimated from randomly sampled cubes whose support did not contain a GT center. The background-subtracted estimate was

$$\widehat{S}_k(\omega) = \max\Big(\overline{R}_{k,\mathrm{GT}}(\omega) - \overline{R}_{k,\mathrm{bg}}(\omega), 0\Big) .$$

The same operation was performed for each class; no ALS first-iteration noise spectrum was used in this direct periodogram experiment.

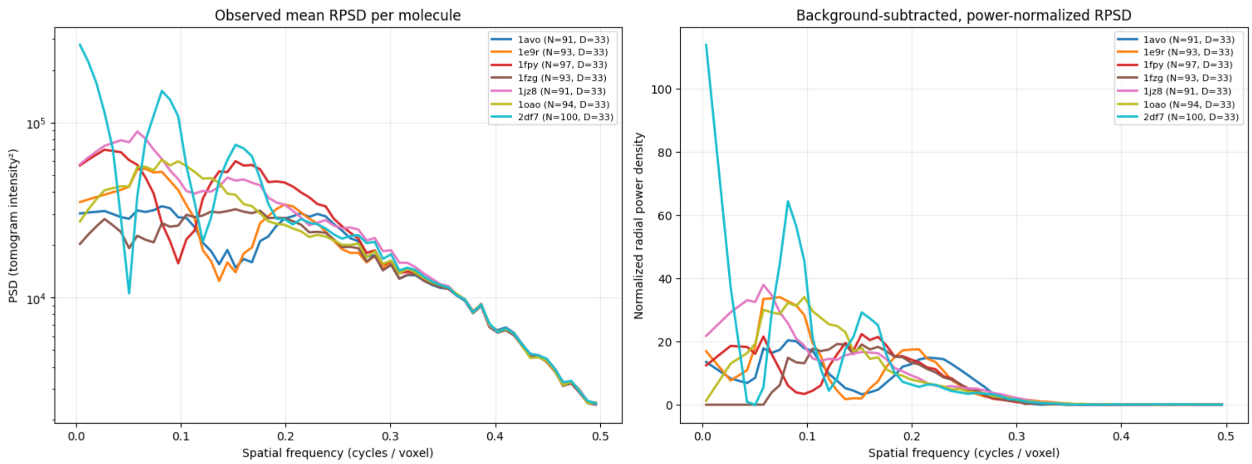


Figure 7: Observed mean RPSD (left) and background-subtracted, power-normalized RPSD (right) for all seven molecules. The largest variation occurs at low frequencies; all curves decay toward zero at high frequencies.

The background-subtracted integrated powers were

k	1AVO	1E9R	1FPY	1FZG	1JZ8	1OAO	2DF7
$\int \widehat{S}_k$	1072.5	1132.5	2179.0	1052.6	2133.5	1507.8	2167.8

in the notebook's calibrated units. Thus 2DF7 is among the strongest classes, but its per-instance spectral integral is not uniquely largest: 1FPY is slightly higher and 1JZ8 is comparable. The likely dominance of 2DF7 must therefore involve the combination of power, occurrence count, spatial extent, and reconstruction response rather than mass alone.

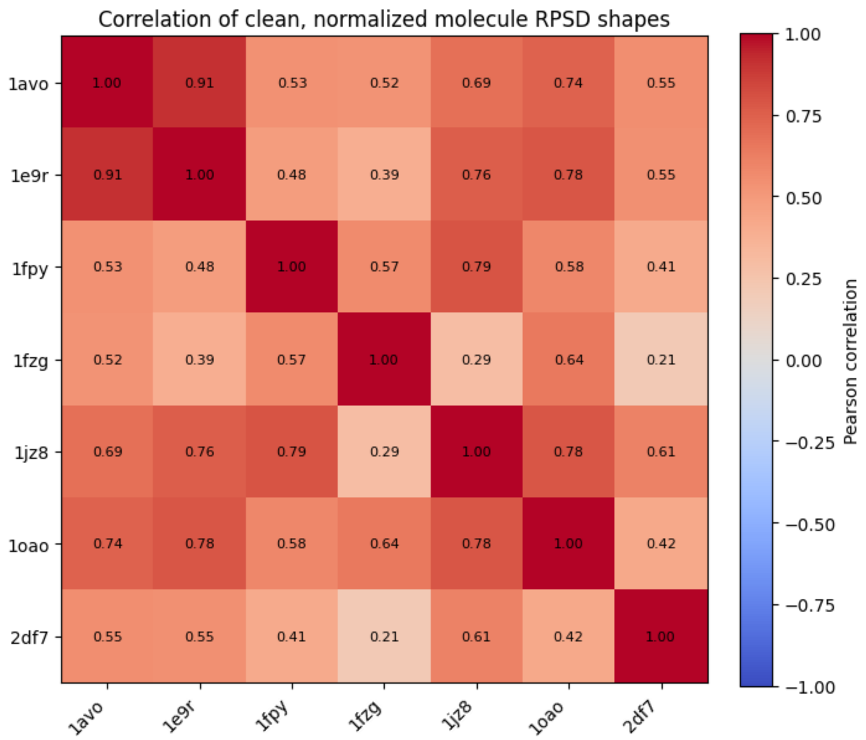


Figure 8: Correlation of the clean, normalized molecule RPSD shapes. Several classes are moderately similar, but the raw low-frequency shapes are not identical. For example, 2DF7 correlates only 0.21 with 1FZG and 0.41 with 1FPY under this direct common-grid comparison.

3.2 Comparison with the recovered KLT spectrum

The first KLT comparison used the original 2DF7-scale settings: $D = 33$, PSD patch 25, two-pass whitening, and the 128^3 ROI. After the whitening and smoothing operations, all reference spectra became much more similar than their direct periodograms. In that baseline comparison the recovered KLT shape had cosine similarity 0.9982 with 1JZ8, 0.9936 with 1FPY, 0.9667 with 2DF7, 0.9799 with the equal-class average, and 0.9690 with the instance-weighted average. This did not provide a stable molecule-specific interpretation.

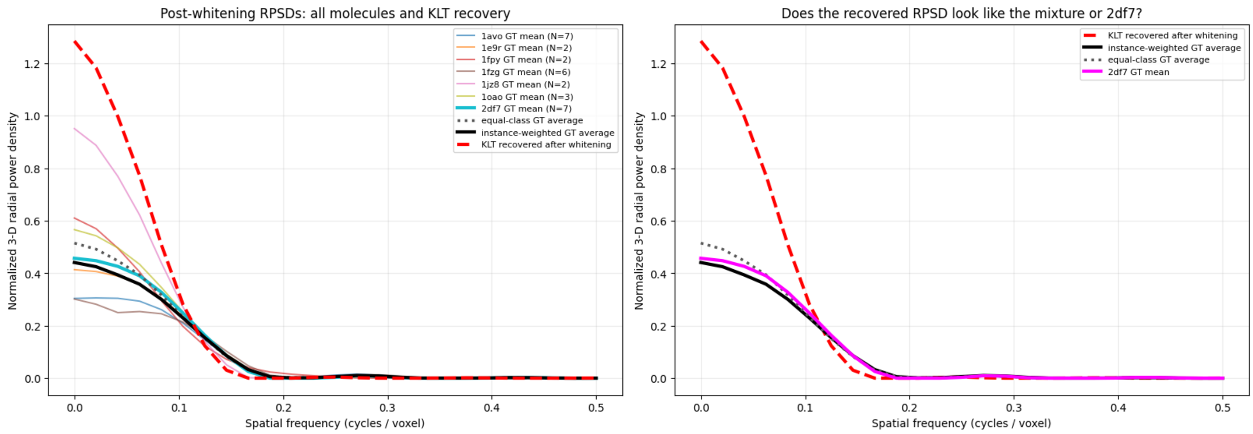


Figure 9: Post-whitening per-molecule spectra and the recovered rank-one KLT RPSD under the original 2DF7-scale settings. Smoothing and common noise subtraction suppress much of the low-frequency variation visible in the direct periodograms.

To make the comparison internally consistent, the experiment was repeated on a common 160^3 ROI with 33^3 patches, two-pass whitening, and the same KLT-compatible autocorrelation estimator for the GT references and the ALS model. The autocorrelation support $d_{\max}$ was swept through $\{7, 10, 12, 16, 24\}$ voxels. The recovered rank-one spectrum agreed most consistently with the pooled, instance-weighted GT mixture:

Table 3: Cosine similarity of the recovered spectrum to selected references in the matched-support experiment.

Reference	$d_{\max} = 7$	10	12	16	24
GT 2DF7	0.9972	0.9627	0.9903	0.9151	0.8539
Equal-class average	0.9967	0.9976	0.9953	0.9972	0.9218
Pooled GT mixture	0.9972	0.9921	0.9989	0.9985	0.9831

At short support all curves are heavily smoothed and nearly indistinguishable. At longer support the recovered spectrum separates from the 2DF7 reference while remaining close to the pooled mixture. Shared background subtraction, clipping at zero, and subsequent normalization can inflate cosine similarity; therefore the trend across support values is more informative than any single near-unity value.

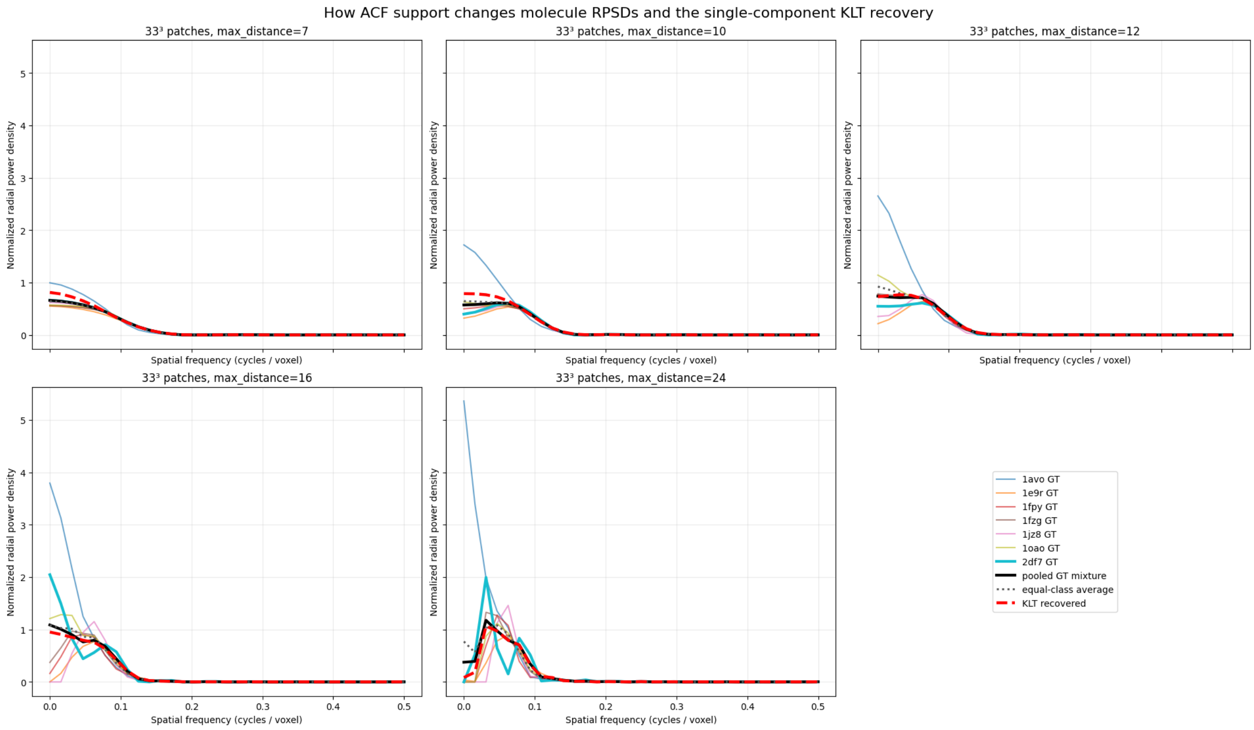


Figure 10: Normalized matched-support RPSDs for the five $d_{\max}$ values. Short support removes much of the discriminating low-frequency structure; longer support exposes larger differences.

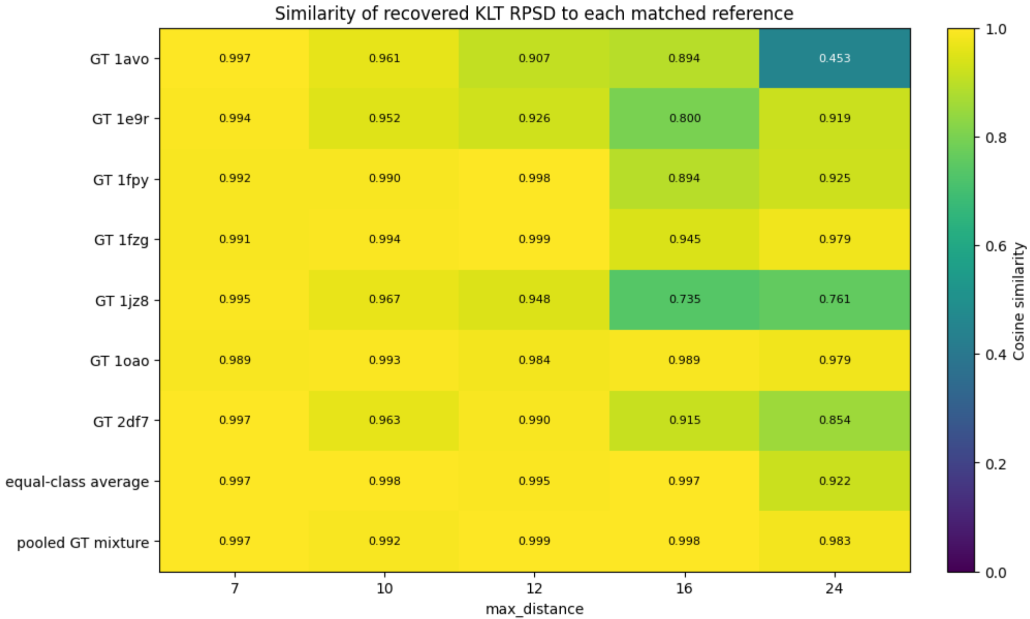


Figure 11: Cosine similarity of the recovered KLT spectrum to each class, the equal-class average, and the pooled instance-weighted mixture.

4 Independent class-scale localization tests

4.1 Why the first cross-class test was insufficient

The first specificity script ran the 2DF7-scale algorithm once and reused its single score tensor. It then changed only the number of retained NMS peaks to match the known count of each class. Because the first eleven peaks were already known to fall on 2DF7, zero recall for the other classes was largely predetermined. That experiment established the composition of one score ranking, but it did not test whether class-specific scale changes alter the learned spectrum or score tensor.

4.2 Fresh fit at every TomoTwin evaluation box

The corrected experiment reran the complete pipeline independently for every class. For class k, the supplied diameter was its TomoTwin evaluation box D_k, and all linked parameters were rescaled:

$$\begin{aligned} p_k &= \text{odd\_floor}(0.8D_k), \\ a_k &= 0.4D_k, \\ r_{\text{NMS},k} &= 0.5D_k, \\ r_{\text{match},k} &= 0.5D_k. \end{aligned}$$

Each run received only its known number of ROI instances N_k. No PDB identity, reference density, molecule-specific RPSD, or molecule-specific template was supplied to ALS or KLT.

Table 4: Target-class recall from seven independent end-to-end fits.

Run	D_k	PSD patch	Known count	Matched	Recall
1AVO	18	13	8	0	0.000
1E9R	21	15	5	0	0.000
1FPY	18	13	4	0	0.000
1FZG	28	21	9	0	0.000
1JZ8	20	15	4	0	0.000
1OAO	20	15	9	1	0.111
2DF7	33	25	12	11	0.917

To identify what the "wrong" runs actually found, every top-N_k list was assigned one-to-one against the union of all molecular GT centers using that run's $D_k/2$ tolerance. The result is striking: almost every scale-specific run still returned mostly 2DF7.

Table 5: GT identity composition of each independently fitted top-N_k list. "None" denotes a pick with no molecular GT match inside the run's tolerance.

Run	D_k	1AVO	1E9R	1FPY	1FZG	1JZ8	1OAO	2DF7	None
1AVO	18	0	0	0	0	1	0	5	2
1E9R	21	0	0	0	0	0	0	4	1
1FPY	18	0	0	0	0	1	0	2	1
1FZG	28	0	0	0	0	0	0	8	1
1JZ8	20	0	0	0	0	0	0	3	1
1OAO	20	0	0	0	0	1	1	6	1
2DF7	33	0	0	0	0	0	0	11	1

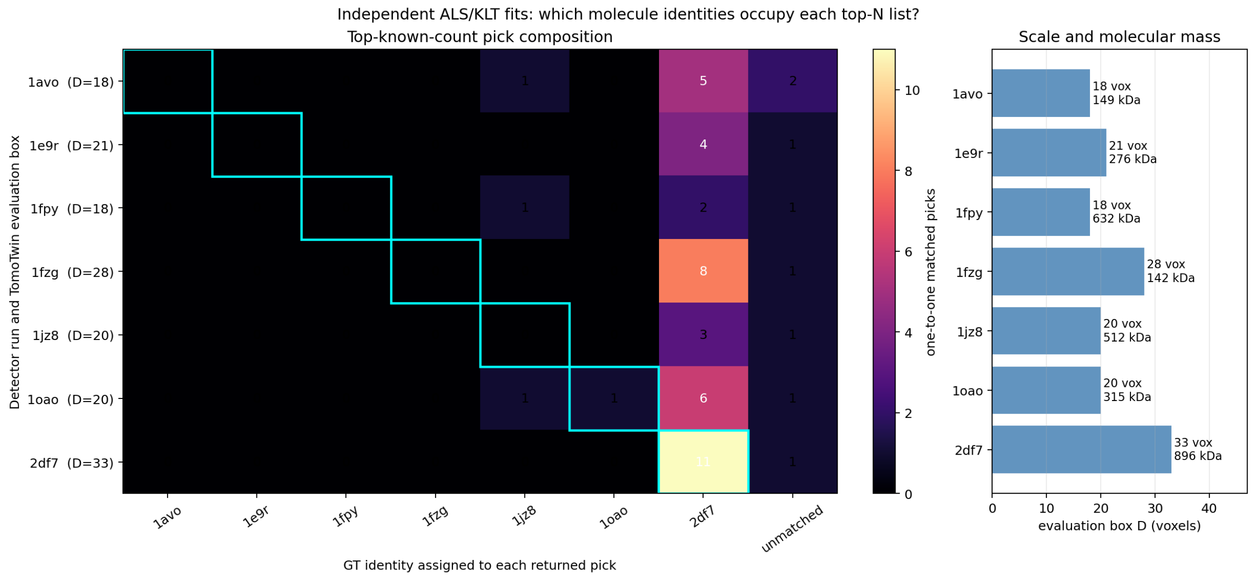


Figure 12: Cross-class composition of the independent known-count runs (left) and the TomoTwin evaluation boxes with molecular masses (right). Cyan outlines mark the requested class identity.

5 One-pass localization at the aggregated particle count

The class-scale tests above probe only the highest N_k peaks of each score ranking. They therefore establish that 2DF7 dominates the top of the ranking, but they do not show whether lower-ranked peaks respond to the other molecules. A missing control was performed by fitting one ordinary rank-one ALS/KLT detector to the untouched raw ROI, fixing all geometry to the largest TomoTwin box $D = 33$, and making one NMS call for the aggregated molecular count

$$N_{\text{all}} = 8 + 5 + 4 + 9 + 4 + 9 + 12 = 51.$$

The NMS radius remained $0.5D = 16.5$ voxels. Picks were assigned globally and one-to-one against all 51 molecular GT centers, with each GT retaining its own class-specific matching tolerance $D_k/2$. GT class labels were not used in ALS, template construction, scoring, or NMS.

The ACF support used in both spectral passes was swept through $d_{\max} \in \{7, 9, 12, 15\}$. Short support performed best (Table 6).

Table 6: One-pass aggregated-count localization with fixed $D = 33$.

$d_{\max}$	Returned	Matched	Unmatched	Recall / precision
7	51	36	15	0.706
9	51	36	15	0.706
12	51	35	16	0.686
15	51	31	20	0.608

At $d_{\max} = 7$, every molecule class contributed at least two matched detections (Fig. 13). The class-level results were 1AVO 4/8, 1E9R 3/5, 1FPY 3/4, 1FZG 6/9, 1JZ8 2/4, 1OAO 6/9, and 2DF7 12/12. The detector therefore recovered all 2DF7 particles while also localizing 24 particles from the other six classes.

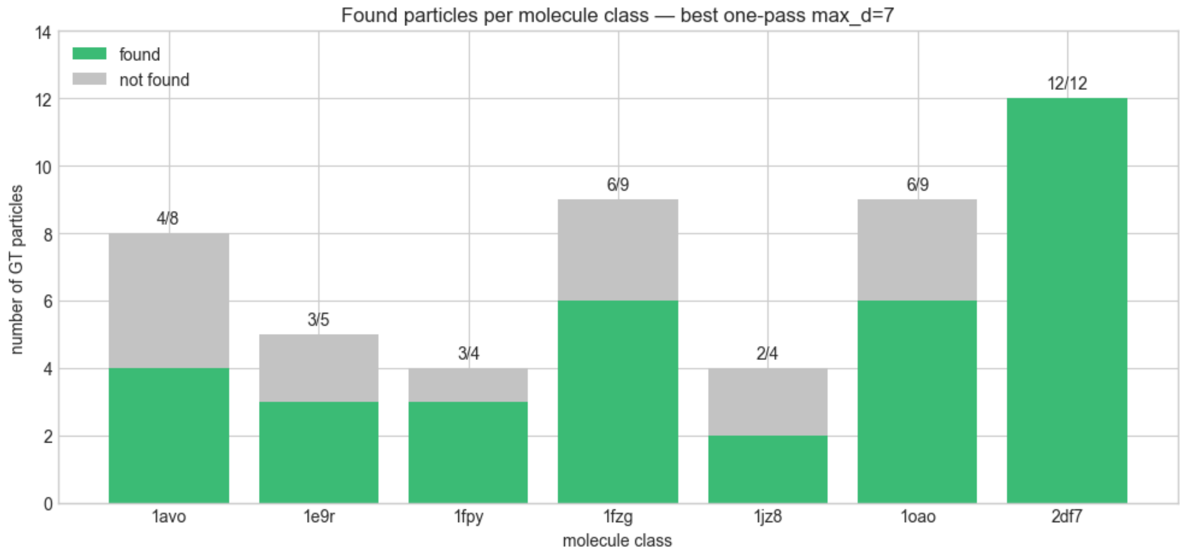


Figure 13: Found and missed particles by molecule class for the best one-pass aggregated-count run ($d_{\max} = 7$). The 36 matched peaks correspond to 70.6% recall; 15 retained peaks had no molecular GT match.

This result changes the interpretation of the earlier specificity tests. The first 12 peaks remain strongly 2DF7-biased, but the complete top-51 list is not 2DF7-only. Rank-one ALS/KLT is therefore useful as a broad particle candidate generator even though its learned RPSD and score ordering are not molecule-specific. Increasing $d_{\max}$ degraded localization, consistent with the higher variance of long-lag ACF estimates from individual noisy patches.

6 Interpretation: the rank-one ALS limitation

Let $R \in \mathbb{R}_+^{J\times F}$ contain the observed patch RPSDs, with one row per patch and one column per radial frequency. The current ALS model is

$$R_{j,f} \approx v_f + A_j\gamma_f, \qquad \text{or} \qquad R \approx \mathbf{1}v^{\mathsf{T}} + A\gamma^{\mathsf{T}}.$$

After subtracting the common background v, the modeled signal matrix is the outer product $A\gamma^{\mathsf{T}}$, whose rank is at most one. Every particle-containing patch must therefore share one spectral shape γ; only its scalar amplitude A_j may change.

In a heterogeneous tomogram, the model cannot retain distinct $\gamma_{1\text{AVO}}, \ldots, \gamma_{2\text{DF7}}$. It learns the single direction that most reduces the total squared reconstruction error. A class's influence is expected to grow roughly with

$$\text{influence}_k \;\propto\; N_k\, \mathbb{E}_{i\in k}\left[\|R_{k,i} - v\|_2^2\right],$$

modulated by patch coverage, missing-wedge distortion, molecular shape, and compatibility with the radial model.

The evidence for a 2DF7 bias at the top of the score ranking is cumulative:

- it is the heaviest class, at 896 kDa;
- it has the largest evaluation box, $D = 33$ voxels;
- it is the most frequent class in the ROI, with 12 centers;
- it has one of the largest measured background-subtracted spectral integrals;
- fresh fits at all seven scales still rank 2DF7 centers highest.

No molecule identity is hard-coded. Diameter changes patch size, Fredholm support, templates, NMS, and tolerance, but it does not tell ALS which molecule should define γ. The class-scale results therefore support the hypothesis that rank-one ALS learns a pooled particle direction whose highest scores favor 2DF7 because of its combined power, extent, and occurrence. The aggregated-count result adds an important qualification: lower-ranked peaks recover all six other classes, so the rank-one detector is not exclusively a 2DF7 detector. These experiments do not establish molecular mass alone as causal.

7 Constrained K-ALS experiment

7.1 Model and one-active-component constraint

The implemented extension replaced the single particle spectrum with $K = 7$ nonnegative spectra:

$$R_{j,f} \approx v_f + \sum_{k=1}^{K} A_{j,k}\Gamma_{f,k}, \qquad \text{or} \qquad R \approx \mathbf{1}v^\top + A\Gamma^\top.$$

Here $\Gamma_{:,k}$ is the k-th recovered particle RPSD and $A_{j,k}$ is its strength in patch j. Under the working assumption that a patch contains at most one centered molecule, each row of A can be constrained by

$$A_{j,k} \geq 0, \qquad \|A_{j,:}\|_0 \leq 1.$$

An all-zero row represents background. This turns the factorization into a combination of nonnegative matrix factorization and hard spectral clustering: each signal patch selects one component, while each component learns its own RPSD.

7.2 Implemented alternating procedure

The experiment used a common 160^3 ROI, 33^3 patches, stride 16, and initially $d_{\max} = 16$. It did not use GT labels or molecular identities during fitting. The implemented steps were:

1. initialize the common background robustly and form nonnegative residual spectra;
2. normalize the residual shapes and use K-means centers as the seven initial component spectra;
3. for every patch and component, compute the best nonnegative scalar projection $a_{j,k}\Gamma_{:,k}$;
4. assign the patch to the single component giving the largest background-relative error reduction, or leave the row inactive;
5. update each component from its assigned, amplitude-normalized residuals, enforce nonnegativity, and renormalize its scale;
6. update the common background from the remaining residual and repeat the constrained assignment and factor updates.

Each recovered $\Gamma_{:,k}$ generated a separate Fredholm/KLT template bank and score tensor. Component identities remained anonymous during fitting. GT was introduced only after scoring, first to map anonymous components to molecule classes and then to measure localization.

7.3 Oracle and shared-diameter evaluations

Two evaluations isolated the effect of component-to-size assignment. In the *oracle geometry* evaluation, each anonymous component was mapped post-hoc to the molecule class with which its score peaks agreed best, and the mapped class's evaluation box was used to construct its templates and NMS. This oracle mapping was not used to fit the spectra. In the *shared-diameter* evaluation, every component instead used the same largest box, $D = 33$. If diameter assignment were the main limitation, the oracle evaluation should have been substantially better.

Table 7: Localization obtained from the constrained $K = 7$ factorization.

Evaluation	Matched / 54	Micro recall	Macro recall	Classes found
Oracle class geometry	6/54	0.111	0.078	2DF7 only
Shared $D = 33$ geometry	6/54	0.111	0.078	2DF7 only

The two evaluations were identical: only six 2DF7 particles were recovered. The post-hoc component-to-class purities were only 0.158–0.333. Repeating the experiment with the original shorter support $d_{\max} = 9$ did not repair the separation; it recovered 5/54, again only 2DF7. Thus the failure at $d_{\max} = 16$ was not solely a long-lag ACF problem.

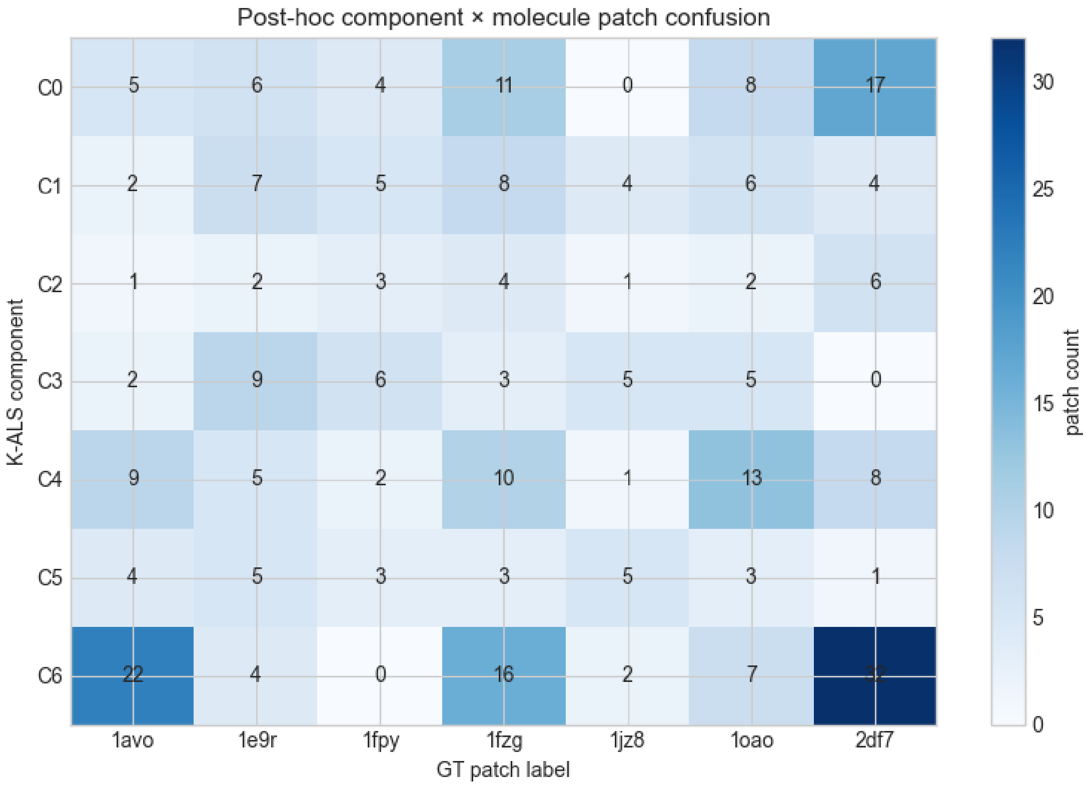


Figure 14: Post-hoc confusion between anonymous K-ALS components and molecular GT classes. Large off-diagonal counts show that the recovered spectral components did not correspond cleanly to molecule identities.

The equal oracle and shared-diameter results show that choosing the correct template size was not the limiting step. The spectral factorization itself did not identify molecule-specific components. Noisy individual patch RPSDs, similar short-support radial shapes, the nonconvex hard assignment, and loss of anisotropic shape information are all consistent with the observed mixing.

8 Clustering RPSDs at detected particle centers

8.1 Detection followed by spectral characterization

The successful aggregated-count run suggests separating localization from identification. The rank-one detector was first run once with $d_{\max} = 7$, $D = 33$, and 51 retained NMS peaks. This fixed set contained 36 GT-matched detections and 15 unmatched candidates. A centered 33^3 patch was then extracted from the whitened volume at each detection, and a new RPSD was estimated from that patch with larger ACF support.

For the initial clustering test, characterization used $d_{\max} = 12$. Each spectrum was converted to a 16-dimensional feature vector: the log-normalized shape of the first 15 radial bins up to 0.45π, plus log total power. Features were standardized and clustered with K-means at $K = 7$. Molecular labels were used only after clustering to find the best cluster-to-class permutation and evaluate the 36 matched detections.

8.2 Direct features versus PCA

Two variants were compared: standardized features directly, and PCA retaining 95% of the variance before K-means. PCA retained six dimensions and 97.3% of the variance. The adjusted Rand index (ARI) measures pairwise agreement after correcting for chance; normalized mutual information (NMI) measures shared information between the cluster and class partitions. Both are label-permutation invariant.

Table 8: Clustering of detected-particle RPSDs at $d_{\max} = 12$.

Features	Mapped accuracy	ARI	NMI	Purity incl. unmatched
Direct standardized	0.444	0.176	0.410	0.412
PCA, 95% variance	0.444	0.176	0.410	0.412

The two methods assigned all 51 candidates identically up to a permutation of cluster labels. PCA therefore did not reveal stronger molecular separation, although repeated-seed tests showed that it stabilized optimization: at $d_{\max} = 12$, mean/minimum clustering stability ARI was 0.981/0.968 with PCA versus 0.937/0.690 without PCA.

8.3 Characterization-support sweep

To distinguish detection support from characterization support, the same 51 detections and the same extracted patches were held fixed while only the second-stage ACF support was swept.

Table 9: Clustering metrics versus RPSD characterization support.

$d_{\max}$	Mapped accuracy	ARI	NMI	Purity incl. unmatched
5	0.389	0.079	0.377	0.392
7	0.417	0.144	0.394	0.431
9	0.417	0.096	0.335	0.451
12	0.444	0.176	0.410	0.412
15	0.417	0.187	0.438	0.451

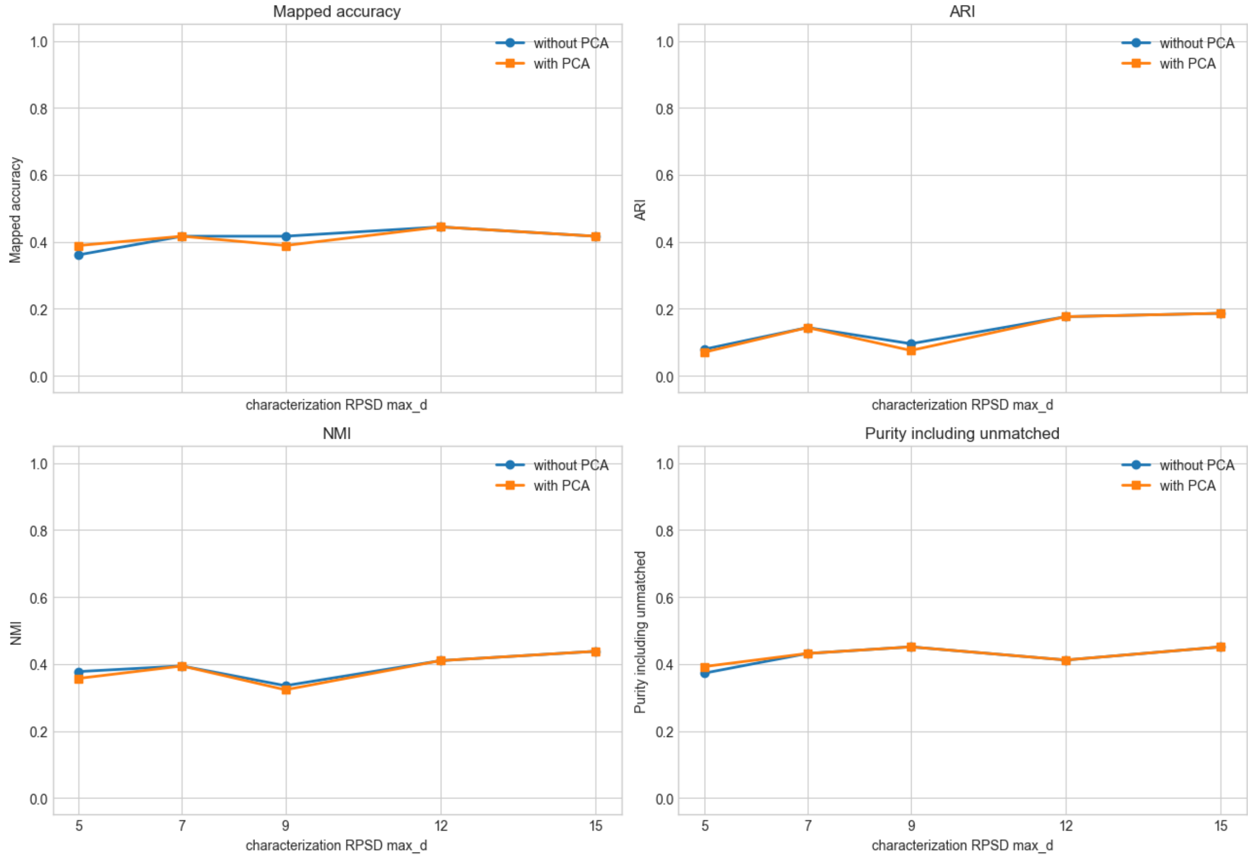


Figure 15: Detected-particle RPSD clustering metrics across characterization support. No single support dominates every metric: $d_{\max} = 12$ gives the highest mapped accuracy, while $d_{\max} = 15$ gives the highest ARI, NMI, and purity.

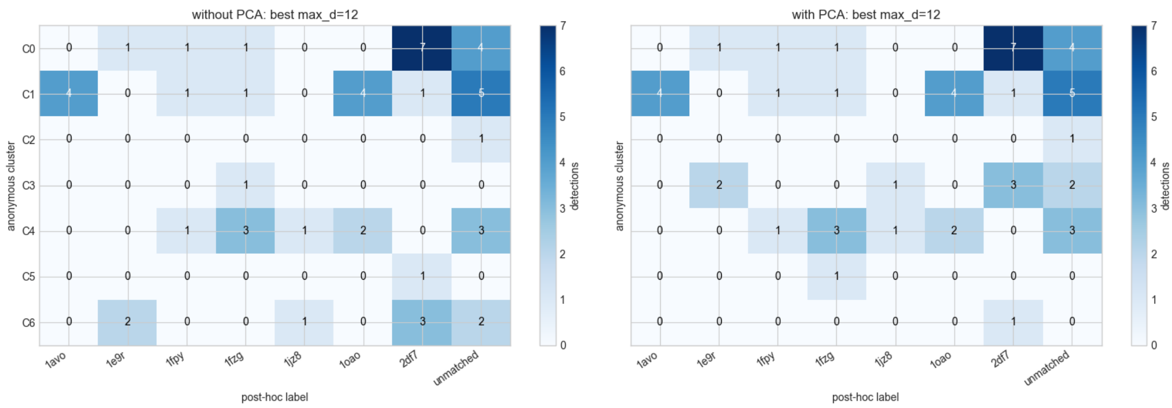


Figure 16: Best post-hoc cluster-to-class confusion matrices from the support sweep. The mixed columns and rows show that detected-patch RPSD alone does not cleanly identify the seven molecule classes.

Larger $d_{\max}$ can expose low-frequency differences that short support smooths away, but it also estimates more noisy long-lag correlations from a single patch. This trade-off explains why $d_{\max} = 7$ is preferable for the initial detector while 12–15 is slightly more informative for characterizing already detected patches. Even at the best support, however, the metrics remain weak: radial spectral content carries some class information but is insufficient for reliable molecular identification.

9 Overall interpretation

The experiments now support a two-part conclusion. First, the current rank-one KLT picker is a useful heterogeneous particle detector: with a common maximum diameter and the aggregate count, it localizes 36/51 particles and represents every class. Its ranking is nevertheless strongly biased toward 2DF7, the largest, heaviest, and most frequent molecule in the ROI.

Second, extracting more spectral components does not by itself solve molecular identification. Joint constrained K-ALS failed before the geometry-assignment stage, and post-detection clustering reached only 44.4% mapped accuracy. PCA improved repeatability but not separation. The remaining gap is therefore not simply an incorrect diameter or a single choice of ACF support; it reflects the limited class specificity of noisy, radially averaged patch spectra.

Data sources

TomoTwin class sizes were taken from `data/boxsizes.json` in the official demonstration dataset: `https://zenodo.org/records/7225386`. Molecular masses and the generalization-tomogram description follow: Rice et al., “TomoTwin: generalized 3D localization of macromolecules in cryo-electron tomograms with structural data mining,” *Nature Methods* (2023), `https://doi.org/10.1038/s41592-023-01878-z`.